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1. Persyaratan Server

Komponen	Minimal
OS	Ubuntu 22.04 / 24.04 LTS
RAM	2 GB
CPU	1 core
Python	3.12+
PostgreSQL	14+
Redis	8+
Git	Terbaru

2. Instalasi Dependencies

```
bash

# Update sistem

sudo apt update && sudo apt upgrade -y


# Install paket yang diperlukan

sudo apt install -y python3 python3-pip python3-venv \
    postgresql postgresql-contrib \
    redis-server git curl nginx
```

3. Setup PostgreSQL Database

```
bash

# Masuk sebagai user postgres

sudo -u postgres psql

sql

-- Buat user database

CREATE USER appuser WITH PASSWORD 'password_anda';


-- Buat database

CREATE DATABASE room_booking_django OWNER appuser;


-- Beri hak akses

ALTER USER appuser CREATEDB;


-- Keluar dari psql

\q
```

4. Setup Redis Server

```
bash

# Konfigurasi Redis

sudo nano /etc/redis/redis.conf
```

Ubah baris:

```
supervised auto → supervised systemd
```

Simpan, lalu:

```
bash
# Restart Redis
sudo systemctl restart redis-server
sudo systemctl enable redis-server
```

```
# Verifikasi
redis-cli ping
# Output: PONG
```

5. Clone Repository dari GitHub

```
bash
# Pindah ke direktori opt
cd /opt

# Clone repository
git clone https://github.com/hackernewbie91/booking-room.git roombooking

# Masuk ke folder project
cd /opt/roombooking
```

6. Setup Virtual Environment & Install Packages

```
bash
# Buat virtual environment
python3 -m venv venv

# Aktifkan
source venv/bin/activate

# Install packages
pip install --upgrade pip
pip install -r requirements.txt
```

7. Konfigurasi Environment (.env)

```
bash
# Buat file .env
nano .env

Isi dengan konfigurasi berikut (sesuaikan):

env

# Django Secret Key (generate random string)
SECRET_KEY=generate-random-string-panjang-50-karakter


# Mode
DEBUG=False


# Allowed Hosts
ALLOWED_HOSTS=domain-anda.com,localhost,127.0.0.1


# Database
DATABASE_URL=postgres://appuser:password_anda@localhost:5432/room_booking_django


# Email (sesuaikan dengan SMTP Anda)
EMAIL_HOST=mail.domain-anda.com
EMAIL_PORT=465
EMAIL_USE_TLS=False
EMAIL_USE_SSL=True
EMAIL_HOST_USER=noreply@domain-anda.com
EMAIL_HOST_PASSWORD=password_email_anda
DEFAULT_FROM_EMAIL=RoomBooking <noreply@domain-anda.com>
```

Simpan dan keluar (Ctrl+O, Enter, Ctrl+X)

8. Jalankan Migrasi Database

```
bash
cd /opt/roombooking
source venv/bin/activate

# Buat tabel di database
python3 manage.py makemigrations bookings
```

```
python3 manage.py migrate
```

Output yang diharapkan:

text

Applying bookings.0001_initial... OK

Applying bookings.0002_... OK

...

9. Buat Superuser Admin

```
bash
```

```
python3 manage.py createsuperuser
```

Isi sesuai petunjuk:

text

Username: admin

Email: admin@domain-anda.com

Password: *****

Password (again): *****

Superuser created successfully.

10. Kumpulkan Static Files

```
bash
```

```
python3 manage.py collectstatic --noinput
```

Output:

text

XXX static files copied to '/opt/roombooking/staticfiles'

11. Setup Daphne Service (Production)

11.1 Buat file service

```
bash
```

```
sudo nano /etc/systemd/system/roombooking.service
```

Isi dengan:

```
ini
```

```
[Unit]
```

```
Description=RoomBooking Django Application (Daphne)
```

```
After=network.target postgresql.service redis-server.service
```

```
Wants=postgresql.service redis-server.service
```

```
[Service]
```

```
User=ubuntu
```

```
Group=ubuntu
```

```
WorkingDirectory=/opt/roombooking
```

```
EnvironmentFile=/opt/roombooking/.env
```

```
ExecStart=/opt/roombooking/venv/bin/daphne -b 0.0.0.0 -p 8880 roombooking.asgi:application
```

```
Restart=always
```

```
RestartSec=5
```

```
[Install]
```

```
WantedBy=multi-user.target
```

Simpan dan keluar.

11.2 Jalankan service

```
bash
```

```
# Reload systemd
```

```
sudo systemctl daemon-reload
```

```
# Enable service (auto-start saat boot)
```

```
sudo systemctl enable roombooking
```

```
# Start service
```

```
sudo systemctl start roombooking
```

```
# Cek status
```

```
sudo systemctl status roombooking
```

```
Harus menampilkan active (running).
```

Selanjutnya tinggal konfigurasi cloudflare tunnel